

BREAK INTO AEROSPACE

Advanced Air Mobility Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Advanced Air Mobility is simultaneously the most discussed and the most misunderstood sector in aerospace engineering. At its broadest, the term encompasses any novel aviation system that moves people or cargo through low-altitude urban and suburban airspace — electric vertical takeoff and landing aircraft (eVTOL), electric short takeoff and landing aircraft (eSTOL), autonomous cargo drones, air taxis operating between vertiports, and the ground infrastructure, airspace management systems, and regulatory frameworks that make any of it work. At its most specific, it is the attempt to build and certify, for the first time, a new category of aircraft that departs and arrives vertically using electric propulsion, carries passengers in a commercial context under Part 135 air carrier authority, and operates with a degree of autonomy that ranges from highly automated to fully autonomous.

The engineering challenges that define this sector are genuinely novel. The aircraft must be quieter than a helicopter at equivalent performance levels — because vertiports in urban centres will face community noise opposition that will terminate operations if the acoustic signature is not dramatically better than the helicopter it is replacing. The propulsion system must be electric — because the operating economics of urban air mobility depend on the per-flight energy cost advantage of electricity over aviation fuel, and because the zero-local-emission requirement for urban operation is increasingly a regulatory condition. The certification must succeed under a regulatory framework that was developed for either fixed-wing aircraft or helicopters, neither of which adequately describes what an eVTOL actually is. And the entire system — aircraft, vertiport, airspace management, booking platform, and passenger experience — must integrate into a commercially viable product that a passenger is willing to pay for and a city is willing to permit.

The employer landscape spans pure-play eVTOL developers who are the closest to commercial service (Joby Aviation, Archer Aviation, Wisk Aero, Lilium's technology lineage, Volocopter, EHang, AutoFlight, Beta Technologies, Supernal, Overair's technology assets), the established aviation industry entering AAM (Airbus CityAirbus NextGen, Boeing subsidiary Wisk, Embraer Eve Air Mobility, Bell Nexus programme, Vertical Aerospace), vertiport infrastructure developers (UrbanV, Skyports, Ferrovia Skyway, Ferrovia, REEF Airports), UTM and airspace management technology companies (NASA UTM programme, Frequentis, Garmin UTM, Wing, Zipline, Airwayz), AAM ecosystem consultancies (McKinsey Center for Future Mobility aviation practice, Roland Berger aviation team, Oliver Wyman aviation), regulatory bodies actively developing AAM policy (EASA, FAA, CAAC, CAA, Eurocontrol), and academic research programmes (Georgia Tech Center for Urban and Regional Air Mobility, Delft Aerospace Faculty, Cranfield AAM research group, MIT IDSS).

The technical baseline is shared across all of these employers — the physics of rotating-wing flight, the electrochemistry of lithium-ion cells, the plasma physics of electric motors — but the specific engineering challenges, the regulatory context, and the career trajectory differ substantially between a propulsion engineer at Joby and an airspace management systems architect at a UTM provider.

PART I: WHAT ADVANCED AIR MOBILITY ACTUALLY LOOKS FOR

By Level

Intern / Graduate

At entry level, AAM employers are not primarily screening for eVTOL-specific knowledge — they are screening for strong fundamentals in the relevant discipline (aerodynamics, structural engineering, electrical engineering, flight dynamics and control, software engineering, or systems engineering), combined with the specific intellectual characteristic that the AAM sector most values: the ability to engage productively with genuinely novel engineering problems for which the textbook does not have an answer.

This last quality is specifically important in AAM because the sector is, by definition, doing things that have not been done before at commercial scale. The handling qualities methodology for an aircraft that transitions between hover and forward flight does not exist in any regulatory document. The certification basis for a distributed electric propulsion system with twelve motors has no precise precedent in Part 25 or Part 27. The urban noise impact assessment for a vehicle that sounds nothing like a helicopter and nothing like a fixed-wing aircraft cannot be directly computed from existing ICAO Annex 16 methods. Candidates who are comfortable engaging with these gaps — who do not reach for the textbook or the handbook when it is not there, but who can reason from physical principles and existing adjacent knowledge toward a technically defensible position — are demonstrating the intellectual character that the sector requires.

Graduate interviews at eVTOL developers typically involve a technical problem relevant to the candidate's specific discipline — a propulsion sizing exercise, a structural loading scenario, a control law design problem — followed by a discussion of how the candidate's approach would change if the standard methodology did not apply to the novel configuration. At UTM and airspace management companies, the screen focuses more on systems thinking, data structures for airspace management, and the candidate's understanding of the regulatory framework within which autonomous aircraft must eventually operate.

Mid-Level (Engineer, Senior Engineer, 3–8 years)

At mid-level, AAM interviews become programme-specific and integration-focused. The sector's defining technical challenge at mid-level is integration — not the design of any individual system, but the management of the interfaces between systems: the propulsion system and the power management system, the flight control system and the degraded-mode propulsion architecture, the aircraft and the vertiport charging infrastructure, the autonomous flight system and the airspace management framework.

The specific question that most reliably reveals mid-level AAM engineering depth is about the failure case: “your twelve-motor eVTOL loses two adjacent motors on the same side during the transition from hover to forward flight at 50 knots — walk me through what happens to the aircraft's attitude and energy state, what the flight control system must do in the next 200 milliseconds, what the pilot sees (if there is a pilot), and what the certification basis requires you to demonstrate about this scenario.” The candidate who can reason through the aerodynamic yaw moment from the asymmetric thrust loss, the flight control allocation problem, the energy state implications for the remaining motors, the certification failure condition classification, and the means of compliance challenge is demonstrating the systems-level AAM engineering depth that hiring managers at eVTOL developers specifically assess.

Senior (Principal Engineer, Technical Lead, Head of Certification)

At senior level, AAM interviews assess the ability to make and defend the architectural and strategic technical decisions that determine whether a programme succeeds. The senior AAM engineer — whether a chief engineer, a head of certification, or a chief systems architect — is the person who decides which novel failure modes are within the existing regulatory framework and which require a new means of compliance; who defines the test matrix that will close the certification evidence package; and who represents the organisation's technical position in the engagement with EASA or the FAA when the authority disagrees.

The distinguishing credential at senior level is having driven a specific certification finding from issue identification through means of compliance development through authority acceptance — a process that in AAM often requires creating the methodology, not just applying it. Engineers who have done this, and who can describe the specific technical argument they made to the authority and how that argument evolved through the engagement, have experience that is unique to the sector's current developmental phase and will become increasingly rare as the methodologies mature and become codified.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

- **SC-VTOL-01 and the FAA powered-lift certification pathway** — the framework that governs every AAM certification in Europe and the US. EASA's Special Condition for VTOL aircraft (SC-VTOL-01,

published 2019 and under continuing development through the CS-VTOL development process) and the FAA's powered-lift category (established by final rule in October 2023, with airworthiness standards based on Part 25/27 with Special Conditions) are the two primary regulatory frameworks for eVTOL certification. Understanding what SC-VTOL-01 requires at the level of specific subparagraphs — CS-VTOL-SC.2000 for the general airworthiness category, SC.2305 for the propulsion system integrity requirement (loss of any single propulsor must leave the aircraft capable of controlled flight and landing), SC.2000 for general categories, SC.2305, SC.2500 for the aircraft systems and equipment requirements, and SC.2510 for the automated systems — is the regulatory literacy that every AAM certification engineer must have. The ELOS (Equivalent Level of Safety) process and the AMC (Acceptable Means of Compliance) development process are the specific regulatory mechanisms through which novel eVTOL certification arguments are progressed, and engineers who have participated in either are demonstrating a specific and rare competency.

- **Distributed electric propulsion failure mode analysis** — the specific FMEA challenge of multi-motor architectures. The safety case for a distributed electric propulsion eVTOL — with six, eight, or twelve individual motors each capable of independent failure — requires a failure mode analysis that is fundamentally different from the twin-engine aircraft FMEA that traditional aviation safety methodology was designed for. The key challenge: each motor failure scenario (single motor, adjacent dual motor, non-adjacent dual motor, single-side triple motor) produces a different aerodynamic and control authority consequence, and the aircraft's certification basis requires demonstration that the worst-case credible failure scenario results in a controlled flight and landing within the certification requirement. Engineers who have constructed an FMEA for a distributed electric propulsion system — who understand what independence assumptions are required between motors on the same bus segment, what common cause failures (a single battery pack supplying multiple motors, a shared control bus) must be excluded from the independence claim, and how the failure consequence is assessed against SC-VTOL-01's controllability requirement — are demonstrating the AAM safety engineering depth that is specifically required at eVTOL developers' certification teams.
- **Hover-to-forward-flight transition aerodynamics and the power required curve.** The transition from hover to forward flight — the flight regime unique to VTOL aircraft — is aerodynamically and controls-mechanically the most complex phase of an eVTOL operation, and the one for which the least regulatory methodology exists. In a pure-lift configuration (multiple rotors providing all lift and thrust), the transition requires simultaneously reducing rotor speed or collective to decrease lift as wing lift builds, increasing forward thrust to accelerate, and managing the resulting pitch, roll, and yaw moments as the aerodynamic authority of the wing increases relative to the rotor authority. In a tilt-rotor or tilt-wing configuration, the transition requires mechanical rotation of the propulsion system that introduces additional load cases and failure modes. The power required during transition — which for many configurations exceeds the hover power requirement due to the combination of rotor-generated thrust and drag in forward flight — is the energy budget constraint that determines the battery sizing and the reserve margins at every point in the mission. Engineers who can sketch the power required curve across the speed range, identify the critical power maximum in transition, and explain what aircraft configuration parameters most strongly influence the transition power hump are demonstrating AAM aerodynamics depth.
- **Urban airspace and UTM** — the U-Space regulatory framework in Europe, UAS Traffic Management in the US, and the conflict of commercial and safety requirements. AAM aircraft do not operate in isolation — they share the low-altitude urban airspace with conventional helicopters, light aircraft, unmanned cargo drones, and eventually thousands of autonomous air taxis simultaneously. The airspace management framework that coordinates these operations — U-Space in Europe (Commission Implementing Regulation EU 2021/664 and its associated regulations), UAS Traffic Management in the US (NASA UTM Research Transition Team output, now implemented through LAANC and evolving toward FAA DroneZone and AAM-specific frameworks) — is the operational infrastructure that determines whether AAM is practically scalable beyond single-aircraft demonstrations. Understanding the U-Space service layers (U1 through U4, with U3 and U4 covering urban operations including manned UAM), what the Network Identification, Geo-Awareness, UAS Flight Authorisation, and Traffic Information services provide, and how the regulatory framework interacts with the commercial requirements of a vertiport operator scheduling dozens of flights per hour are the airspace management competencies that UTM system

architects and AAM ecosystem engineers require.

- **Vertiport design and infrastructure** — the physical integration point between the aircraft and the city. The vertiport is the critical infrastructure element that determines where AAM can operate and how efficiently. Its design requirements are driven simultaneously by the aircraft's geometric and performance constraints (the Final Approach and Takeoff Area geometry, the obstacle limitation surface, the ground effect on approach, the blast and noise impact on surrounding infrastructure), the passenger experience requirements (throughput, dwell time, security, connection to other transport modes), the power infrastructure requirements (the charging system that must recharge the aircraft battery between flights in the minimum possible time while staying within the grid connection capacity), and the regulatory framework (EASA's Vertiport Design Standards under CM-AD-005, ICAO Annex 14 Volume II which is being updated for vertiports). Engineers who understand the multi-constraint design problem of a vertiport — specifically the trade between FATO size (which determines the aircraft payload-range capability at that site) and urban real estate cost — are demonstrating the AAM infrastructure systems engineering depth that vertiport developers and AAM operators require.
- **Handling qualities for eVTOL** — why existing standards do not apply and what the emerging methodology requires. Military handling qualities standards (ADS-33E-PRF for rotorcraft, MIL-HDBK-1797A for fixed-wing) were developed for pilots with training backgrounds in conventional aircraft. The Cooper-Harper Rating Scale — the universal subjective handling qualities tool — assumes a human pilot applying continuous corrective control inputs. Neither of these frameworks maps cleanly to an eVTOL that transitions between hover and forward flight, that may have limited or no conventional control surfaces, that may be operated by a pilot with fixed-wing rather than rotorcraft training, or that is intended for eventual single-pilot or pilotless operation. The specific handling qualities challenges for eVTOL in the transition regime — what happens to pilot workload when the rotor-dominated control authority of hover transitions to the wing-dominated authority of forward flight, what the pilot's instrument scan priority is during the transition, and what the minimum acceptable handling qualities are for a pilot qualified under Part 135 but not helicopter-rated — are the AAM-specific handling qualities questions that flight dynamics and certification engineers must engage with, and for which the answers are being developed through ongoing research and regulatory dialogue.
- **Battery certification for aviation** — the specific means of compliance challenges and the current state of EASA and FAA guidance. As covered in detail in the Electric & Hybrid-Electric Aviation guide, the certification of lithium-ion battery packs for aviation use presents specific and largely novel means of compliance challenges: the absence of in-service failure rate data for aviation-grade cells, the thermal runaway containment demonstration, the state-of-health monitoring accuracy requirement, and the operational life limit derivation methodology. For AAM specifically, these challenges are compounded by the operational intensity of the intended use — an air taxi battery that is charged and discharged multiple times per day, every day, at high C-rates, will degrade substantially faster than a heritage GEO satellite battery cycled once per day at low C-rate. The certification basis for the resulting degradation — what minimum capacity retention the battery must demonstrate at end of approved life, and how this is demonstrated through accelerated life testing — is one of the most actively negotiated aspects of current AAM certification programmes.
- **Noise** — the community acceptance constraint that determines where vertiports can be built. Urban community noise is the non-technical constraint most likely to determine AAM's commercial viability in any specific city. A vertiport operating in a dense urban neighbourhood that generates noise complaints will face regulatory action that can shut down operations before the business model has time to mature. The noise sources specific to eVTOL — rotor blade vortex interaction (BVI) noise, blade passing frequency tone noise, electric motor electromagnetic noise, and the broadband turbulence ingestion noise from multi-rotor configurations — produce a noise signature that is different in character from both helicopter noise (BVI-dominated, impulsive) and fixed-wing noise (jet and airframe dominated). Understanding what acoustic metrics are relevant for eVTOL community noise assessment (Lden for long-term exposure, Lmax for individual event annoyance, and the proposed SENEL metric for operational noise certification), what the ICAO Annex 16 Volume I coverage for eVTOL currently provides (limited, with ongoing CAEP working group development), and what the community noise engagement process requires from a vertiport developer (noise modelling, community consultation, noise action plan) are the

AAM noise engineering and policy competencies that certification, operations, and business development roles require.

- **Autonomy and the regulatory pathway** — the current limits of autonomous operation approval and the technical requirements for higher levels. All currently certifying eVTOL aircraft are being certified with a pilot on board — not because the aircraft requires one, but because the regulatory framework for fully autonomous commercial passenger transport does not yet exist. The progression toward autonomous or remote-piloted operation — the commercial goal for AAM economics, because eliminating the pilot cost dramatically changes the unit economics — requires both a technical solution (the autonomous flight management system that can handle all normal operations and a defined envelope of abnormal operations without pilot intervention) and a regulatory pathway (a certification basis for an autonomous system that manages the aircraft's safety case without a certificated pilot in command). Understanding where the current regulatory boundary is (Part 135 requires a pilot in command; BVLOS drone operations have their own evolving framework; fully autonomous passenger-carrying commercial operations have no established US or European regulatory pathway), what the technical requirements for autonomous safety are (the ASTM F3269 Means of Compliance for detect-and-avoid, the DO-178C software assurance requirements for the autonomous flight management function), and what the industry's engagement with the FAA and EASA on this topic looks like are the autonomy regulatory competencies that AAM systems engineers and certification specialists require.
- **Commercial model and ecosystem** — the route economics, vertiport throughput, and the demand model. AAM is not merely an engineering challenge — it is a commercial challenge, and engineers working in the sector who do not understand the commercial constraints will make engineering decisions that are technically sound but commercially disqualifying. The key commercial parameters: the seat-mile cost that makes an air taxi competitive with ground transport for time-sensitive users (approximately \$3–5/mile is the target for initial operations, with a long-term target of \$0.50–1.00/mile for mass-market adoption); the fleet utilisation rate that makes the aircraft economics work (typically 4–6 revenue flights per aircraft per day in initial operations, scaling to 12–15 as vertiport infrastructure matures); and the vertiport throughput that determines whether a specific route can be served (a vertiport with a single FATO and a 10-minute turnaround cycle can serve 6 aircraft per hour, which limits the revenue flight frequency on a route). Engineers who understand these commercial parameters — and who design aircraft, vertiports, and airspace management systems with the commercial constraint explicitly in mind — are more valuable to the businesses that are trying to make AAM viable than engineers who optimise for engineering excellence without regard to commercial consequence.

How AAM Engineering Interviews Differ from Conventional Aviation and from Generic Aerospace

- **The absence of established methodology is a feature, not a bug.** In most aerospace engineering disciplines, the interview probes whether you know the established method and can apply it correctly. In AAM, the interview often probes whether you can develop a credible method when the established one does not exist. “How would you assess the handling qualities of an eVTOL aircraft in the transition regime?” does not have a handbook answer — the candidate who says “I would use ADS-33E” is demonstrating that they have not engaged with why ADS-33E does not apply to a 12-motor electric aircraft transitioning from hover to 150 knots. The candidate who says “I would start from the mission task elements relevant to eVTOL operations, define the aircraft configuration-dependent performance demands of each, and develop a pilot-in-the-loop simulation campaign to collect Cooper-Harper ratings as a function of configuration — then use those ratings to derive bandwidth and phase delay requirements specific to the eVTOL flight regime” is demonstrating the methodological creativity that the sector requires.
- **The multi-domain integration question is specifically assessed.** AAM is not an aerodynamics problem, or a propulsion problem, or a certification problem, or an airspace problem — it is all of these simultaneously, with each domain's requirements imposing constraints on the others. The interview specifically probes the candidate's ability to reason across these domain boundaries: what the noise constraint imposed by the community acceptance requirement means for the rotor blade design, which means for the hover power required, which means for the battery sizing, which means for the aircraft payload capacity, which means for the route economics that must ultimately justify the vertiport infrastructure investment. Candidates who can trace this chain of constraint propagation — who think in terms of system-level im-

pacts of single-domain decisions — are demonstrating the systems engineering perspective that AAM specifically values above single-domain technical excellence.

- **The regulatory engagement is unusually direct compared to other aviation disciplines.** In conventional aviation, the regulatory framework is largely settled — CS-25 has been developed over decades, and the engineer’s job is to demonstrate compliance. In AAM, the regulatory framework is being developed in parallel with the aircraft, and the industry’s technical engagement with EASA and the FAA is direct, frequent, and technically substantive. AAM certification engineers attend the same CAEP working groups that are writing the standards they will eventually certify against. Senior engineers from Joby, Archer, and Wisk participate directly in FAA Aviation Rulemaking Advisory Committee meetings on powered-lift standards. This direct regulatory engagement — writing G-1 issue papers, responding to authority questions with engineering data, proposing means of compliance for novel failure modes — is a specific and valued competency that engineers from conventional aviation programmes rarely develop.
- **The failure of multiple high-profile programmes is part of the technical knowledge base.** Lilium’s insolvency in late 2024, Overair’s shutdown, and the technical difficulties of several other programmes are not merely business failures — they are engineering case studies that the AAM community has studied carefully. Understanding why the Lilium Jet’s ducted fan configuration produced a less favourable transition power curve than the published projections, what the certification challenges of a novel propulsion architecture with no heritage contributed to the programme’s difficulties, or what the thermal management challenge of a high cell-count battery pack at high discharge rates means for a specific vehicle class are the engineering lessons from programme failures that interviewers in the AAM sector use as case studies. Candidates who engage with these failures as engineering content — rather than as business news — are demonstrating the sector-engaged technical curiosity that AAM employers value.
- **Time-to-certification is a competitive survival constraint.** In conventional aerospace, programmes that run over schedule are expensive and embarrassing. In AAM, programmes that run significantly over schedule run out of funding before they can generate revenue. The commercial viability of every AAM developer depends on completing certification before investors lose confidence, and the engineering decisions that affect the certification timeline — the complexity of the certification basis, the number of novel means of compliance required, the test matrix size — are strategic decisions as much as technical ones. Senior engineers who understand how the novelty of a technical approach affects the certification schedule — who can reason about whether a novel propulsion architecture is worth the certification time cost relative to a more conventional approach with shorter certification uncertainty — are demonstrating the engineering-commercial integration that the sector’s leadership requires.

PART II: 20 SPECIFIC PREP TASKS

INTERN / GRADUATE

1. Understand the AAM ecosystem — the aircraft, the vertiport, the UTM, the operator, and how they interconnect 4–5 hours

Skill: The system-level map of the entire industry that frames every engineering role within it.

Why the field cares: AAM is a system of systems — the aircraft alone does not produce a commercially viable service without the vertiport infrastructure to support it, the UTM framework to manage its airspace, the regulatory approval to carry passengers commercially, and the booking platform to connect passengers to the service. A candidate who understands only the aircraft engineering is missing the context within which every aircraft engineering decision is made. The battery designer who does not understand that the charging infrastructure at the vertiport is limited by the grid connection capacity cannot make the right trade-off between charging speed and battery cycle life. The flight dynamics engineer who does not understand that the vertiport FATO geometry constrains the aircraft’s hover performance cannot correctly specify the hover thrust margin. Understanding the

4–5 hours

ecosystem — not as background information but as the constraint environment within which every technical decision is evaluated — is the AAM engineering mindset.

Done signal: You can describe the AAM ecosystem's five major components — aircraft, vertiport, UTM/airspace management, operator/airline, and regulatory framework — and for each, state what it must provide to the system, what its primary engineering constraints are, and one specific interaction between it and each of the other components. You can explain why a change in the vertiport's charging infrastructure power limit (say, from 500kW to 200kW per pad) would cascade through the aircraft design (battery capacity versus charge time trade), the operator schedule (minimum ground time), and the route economics (revenue flights per day).

5–6 hours

2. Understand SC-VTOL-01 and the FAA powered-lift category — the certification framework at the level of its specific requirements

Skill: The regulatory foundation that governs every AAM aircraft certification in Europe and the US.

Why the field cares: SC-VTOL-01 and the FAA powered-lift final rule are the governing regulatory documents for eVTOL certification, and understanding them at the level of their specific requirements — not just their existence — is what distinguishes engineers who can contribute to a certification programme from those who cannot. The key provisions: the three performance-based categories (Basic, Enhanced, and Enhanced+) and what each requires for the loss of a single propulsive unit; the propulsion system integrity requirement that the loss of any single propulsor must leave the aircraft capable of controlled flight and landing; the automated systems requirements that govern the design and certification of flight control automation; and the novel means of compliance process through which the authority accepts technically equivalent demonstrations for requirements that cannot be met by the prescribed standard method. Reading SC-VTOL-01 — not a summary of it, the document itself — is a minimum preparation step for any AAM engineering candidate, and the interview will reveal whether this has been done.

Done signal: You have read SC-VTOL-01 (available on the EASA website as a published Special Condition) and can describe without consulting the document: the three performance categories and their key differences, what CS-VTOL-SC.2305 requires for propulsion system integrity, what the automated systems requirements in SC.2510 specify, and what the ELOS process is and how it is used for novel means of compliance. You can explain one specific requirement from SC-VTOL-01 that does not directly map to CS-25 or CS-27, and describe why the difference reflects the novel flight characteristics of eVTOL aircraft.

4–5 hours

3. Understand the power required curve for an eVTOL — hover power, transition power, and cruise power — and what determines each

Skill: The aerodynamics of the eVTOL mission profile, connecting aircraft design to battery sizing.

Why the field cares: The power required curve — power as a function of airspeed, from hover through transition to cruise — is the fundamental aerodynamic characterisation of an eVTOL aircraft, and it drives the battery sizing, the motor rating, the inverter peak power requirement, and ultimately the aircraft's payload-range capability. For a helicopter-like multicopter, hover power is typically the peak requirement (induced power from rotor thrust dominates). For a lift-plus-cruise configuration (dedicated hover rotors plus a cruise propulsor and fixed wing), the transition may produce a power maximum higher than hover because the lifting rotors are windmilling in forward flight while the wing has not yet fully taken over. Understanding which configuration produces which power curve shape, why the transition power hump occurs in lift-plus-cruise configurations, what design parameters most strongly reduce the hump (rotor folding, variable-pitch blades, rotor disk loading), and what the battery sizing implication is of a 30-second transition at 20% above average cruise power are the eVTOL aerodynamics fundamentals that every AAM engineer must understand regardless of their specific discipline.

Done signal: You can sketch the power required versus airspeed curve for a lift-plus-cruise eVTOL configuration and a tiltrotor configuration, identify the transition power hump for the lift-plus-cruise

4–5 hours

case, and explain physically why it occurs. You can explain what disk loading means and why lower disk loading reduces hover power at the cost of larger rotor diameter and therefore larger aircraft footprint and higher rotor noise. You can compute from actuator disk theory the hover power required for a specified aircraft all-up weight and rotor disk area, and reason about how this scales with aircraft size.

2–3 hours

4. Prepare behavioural answers demonstrating systems thinking, comfort with regulatory ambiguity, and engagement with programme failure as a learning resource

Skill: The character signals that AAM employers specifically probe.

Why the field cares: AAM employers — particularly at the leading certification-phase developers — assess whether candidates can engage productively with ambiguity, whether they take a systems rather than a single-domain perspective, and whether they treat programme failures as engineering information rather than as unrelated business events. The stories that are weakest in AAM interviews demonstrate deep domain expertise applied in isolation (“I optimised the motor efficiency by 3%”). The strongest demonstrate cross-domain thinking under uncertainty (“I identified that the motor efficiency improvement increased peak current at the transition operating point, which required a revision of the battery pack protection threshold, which in turn required re-negotiating the SC-VTOL-01 means of compliance for the propulsion system integrity demonstration”). The candidate who has engaged with publicly available information about AAM programme challenges — and who can discuss them as engineering case studies rather than as business news — is demonstrating the sector-engaged mindset that AAM employers look for.

Done signal: Four STAR stories covering: a situation where you identified a cross-system implication of a single-system decision; a time you developed a methodology or approach when the standard one did not apply; a situation where you engaged constructively with regulatory ambiguity rather than waiting for clarity; and a time you learned from a project or programme failure — yours or someone else’s — and applied that learning to a subsequent decision. At least two stories involve specific technical content from your discipline that had a cross-system consequence.

3–4 hours

5. Understand urban noise and community acceptance — what eVTOL sounds like, why it matters commercially, and what the acoustic engineering challenge is

Skill: The community acceptance constraint that determines where AAM can operate.

Why the field cares: Community noise is, for many AAM developers, the most important non-technical constraint on their commercial viability — more immediately constraining than the certification timeline or the battery specific energy. A vertiport in an urban residential area that generates community complaints will face municipal regulatory action that can prevent operations regardless of the aircraft’s airworthiness certification. Understanding what eVTOL noise sounds like to a community listener — the tonal character of the blade passing frequency harmonics, the impulsive character of BVI noise during approach descent, the broadband turbulence noise from multiple interacting rotors — and why this character matters for annoyance beyond the simple dB(A) level (tonal noise at a specific frequency is disproportionately annoying relative to its SPL; impulsive noise perceived as intrusive at levels that would be acceptable as steady noise) is the acoustic awareness that every AAM engineer who interacts with operations, certification, or business development must have.

Done signal: You can explain what blade vortex interaction (BVI) noise is, why it is produced during descent and deceleration, and why it is associated with the characteristic impulsive “slap” of helicopter noise that is disproportionately annoying to communities. You can describe what tonal noise from a multi-rotor eVTOL configuration sounds like — why multiple rotors at the same speed produce a more intrusive tone than a single rotor at the same total thrust — and explain what design approaches reduce tonal noise (variable rotor speed, rotor phasing, blade number selection). You can describe what acoustic metrics are relevant for community impact assessment (L_{den} for long-term exposure, L_{Amax} for single-event annoyance) and why the SENEL metric has been proposed

specifically for eVTOL operations.

3–4 hours

6. Get hands-on with the AAM technical literature — read five recent published papers on a specific AAM technical challenge relevant to your discipline

6–8 hours

Skill: Specific technical engagement with the field's active research frontier.

Why the field cares: Unlike conventional aviation, where the technical knowledge is primarily codified in textbooks and regulatory documents, AAM's most current knowledge exists in the research literature — AIAA SciTech and AIAA Aviation conference proceedings, NASA Technical Reports on Urban Air Mobility, EASA research publications on VTOL handling qualities and noise, and the SAE International Journal of Aerospace. Candidates who have read recent technical papers on the specific challenge most relevant to their discipline — handling qualities in the transition regime for flight dynamics candidates, thermal runaway containment for battery engineers, vertiport FATO geometry optimisation for infrastructure engineers, or UTM conflict resolution algorithms for software engineers — are demonstrating the specific technical engagement that generic “passion for aviation” statements cannot substitute. The AIAA papers are behind a paywall but often available through university library access, and NASA Technical Reports are freely available.

Done signal: You have read at least five recent (2021–2025) technical papers on one specific AAM engineering challenge relevant to your discipline — naming the papers, their authors and organisations, the key technical finding of each, and one specific technical question that each paper raises but does not fully answer. You can discuss these papers as engineering content — not as bibliographic references — in an interview conversation about the specific technical challenge they address.

MID-LEVEL (~3–8 years)

All graduate tasks still apply. Technical stories must come from professional or serious research contexts. Regulatory knowledge must extend to specific subparagraphs, not just the framework level.

7. Build thorough working knowledge of the distributed electric propulsion safety case — FMEA structure, independence assumptions, and the SC-VTOL-01 controllability requirement

5–6 hours

Skill: The safety engineering methodology for the propulsion architecture that defines eVTOL.

Why the field cares: The safety case for a distributed electric propulsion eVTOL is the most technically novel aspect of AAM certification. Traditional twin-engine aircraft safety analysis establishes that the loss of one engine leaves the aircraft capable of continued flight with reduced performance. For an eVTOL with twelve motors, the failure space is vastly more complex: any subset of motors can fail simultaneously through a common cause (a shared bus, a common software function, a single battery pack), and the aerodynamic consequence of each failure combination is specific to the geometry and depends on the flight phase at the time of failure. The FMEA must address every credible failure combination, establish the independence assumptions that allow low-probability combinations to be excluded from the worst-case analysis, and demonstrate that the worst-case credible failure scenario results in a controlled landing within the SC-VTOL-01 performance category limits. Engineers who have constructed an FMEA of this type — who can describe the independence argument for motors on separate bus segments, the common cause failure modes that must be explicitly considered, and the flight control allocation that manages the worst-case single failure — are demonstrating the AAM safety engineering depth that is specifically required.

Done signal: You can describe the structure of an FMEA for a distributed electric propulsion eVTOL — identifying the top-level functional failure condition (loss of all lift), the intermediate failure modes (loss of multiple motor groups), the basic failure modes (single motor failure, bus failure, battery contactor failure), and the independence assumptions required at each level. You can explain what SC-VTOL-01's Category Enhanced requirement means for the worst-case single propulsor loss, and describe what “controlled flight and landing” means in the context of an aircraft with eight motors losing one — specifically what the degraded performance envelope must still include. You

can identify one common cause failure mode that would invalidate the independence argument for motors on adjacent bus segments, and describe what design feature prevents it.

5–6 hours

8. Understand the vertiport design requirements — the FATO, the obstacle limitation surface, the charging infrastructure, and the passenger throughput constraint

4–5 hours

Skill: The multi-constraint infrastructure design challenge at the heart of AAM commercial viability.

Why the field cares: The vertiport is not merely a place where the aircraft lands — it is the infrastructure element that determines where AAM can operate, how many passengers can be served per hour, how quickly the aircraft can be turned around, and therefore whether the route economics work. EASA's Vertiport Design Standards (CM-AD-005, 2022) and ICAO Annex 14 Volume II (under revision to include vertiport provisions) define the regulatory geometry requirements: the FATO (Final Approach and Takeoff Area) must be large enough for the aircraft's approach and departure geometry, the obstacle limitation surface must be clear of obstructions that would prevent a safe go-around, and the TLOF (Touchdown and Liftoff Area) must withstand the aircraft's maximum landing weight with a defined load distribution. The charging infrastructure must provide sufficient power to charge the battery from minimum reserve to operational charge in the minimum ground time, within the site's electrical grid connection capacity. Understanding how these constraints interact — how the FATO size affects the aircraft size (and therefore payload), how the grid connection limit affects the charging time (and therefore minimum ground time), and how the passenger processing time drives the total turnaround cycle — is the systems-level vertiport engineering that infrastructure developers and AAM operators require.

Done signal: You can describe the key geometric requirements for a vertiport FATO — its minimum dimensions relative to the aircraft's rotor diameter or overall dimensions, the obstacle limitation surface gradient, and the separation requirement between simultaneous FATO operations. You can explain how the FATO geometry constrains the aircraft's approach and departure procedure, and what the consequence is of a steeper approach path for the community noise footprint. You can describe the charging infrastructure sizing problem — computing the peak charging power required to charge a specified battery capacity in a specified ground time, and assessing whether the required power exceeds a typical urban grid connection capacity.

9. Prepare a technically specific story about developing or applying a novel engineering methodology — one where the standard approach did not apply

2 hours

Skill: The methodological creativity that is the defining competency of AAM engineering.

Why the field cares: The AAM sector is defined by the requirement to develop new engineering methods — not to apply existing ones. The handling qualities methodology for eVTOL in transition did not exist before 2018. The thermal runaway containment demonstration methodology for aviation battery packs did not have an established EASA-accepted means of compliance before 2020. The vertiport noise model for a 12-rotor eVTOL had no precedent in ICAO Annex 16. Engineers who have developed or contributed to a novel methodology — even in a research or academic context — and who can describe the development process (the gap in the existing methodology, the analytical or experimental approach taken to fill it, the validation of the new method, and the sensitivity to key assumptions) are demonstrating the intellectual competency that the sector most values and that is hardest to find.

Done signal: You have a story about developing or applying a novel methodology — in a professional or serious research context — where the standard approach was not directly applicable to the problem. The story describes the specific gap in the existing methodology, the approach you took to develop an alternative (citing any literature, physical principles, or adjacent methodologies you drew on), the validation step you applied to assess the method's reliability, and the key assumptions that the method depends on. The story is specific — it names the problem, the gap, and the approach — not generic.

4–5 hours

10. Understand the UAC airspace architecture — U-Space in Europe, UTM in the US, and the specific conflict management and deconfliction services

Skill: The airspace management framework that determines how multiple AAM aircraft share the low-altitude urban sky.

Why the field cares: A single eVTOL operating under visual flight rules in uncongested airspace does not require sophisticated airspace management. Dozens of eVTOL aircraft, unmanned cargo drones, conventional helicopters, and news helicopters sharing the airspace over a dense urban area — the commercial scenario that makes AAM economically viable — requires a systematic airspace management framework that allocates routes, resolves conflicts, separates aircraft, and responds to emergencies in real time. U-Space (EU 2021/664) defines four service layers: U1 (e-registration, geo-awareness), U2 (flight authorisation, tactical deconfliction, network identification), U3 (dynamic capacity management, strategic conflict resolution), and U4 (real-time monitoring, incident response for extreme density operations). Understanding what service each layer provides, who is responsible for providing it (U-Space Service Providers — USSPs — in a competitive market), and how the system handles a scenario where two aircraft are converging on the same vertiport simultaneously is the airspace architecture knowledge that UTM system engineers and AAM operations planners require.

Done signal: You can describe the four U-Space service layers and what each provides operationally. You can explain the role of the Common Information Service (CIS) — the regulatory infrastructure that USSPs connect to for airspace data sharing — and why it is a regulatory function rather than a commercial function. You can describe what a tactical deconfliction service does when it detects two aircraft on converging trajectories — the alert threshold, the conflict resolution authority, and the pilot or autonomous system's obligation in response. You can compare the U-Space approach with the FAA's UTM framework and identify one significant architectural difference.

4–5 hours

11. Understand the Part 135 commercial operation requirements for AAM — what the operator needs to be certificated and how this differs from Part 121

Skill: The commercial operating regulatory framework that determines how AAM services are delivered.

Why the field cares: An eVTOL aircraft with a type certificate cannot carry fare-paying passengers unless the operator also holds an Air Carrier Certificate under 14 CFR Part 135 (for the US) or an Air Operator Certificate under EASA Air Ops (for Europe). These regulatory frameworks impose requirements on the operator that go beyond the airworthiness requirements of the type certificate: the operator must have an approved Operations Manual, an approved Maintenance Programme, a Safety Management System, a Flight Operations Quality Assurance programme, and a training curriculum for its pilots and crew members. The specific challenge for AAM: Part 135 was written for existing aircraft categories (small aircraft, multi-engine piston, turboprop, jet), and its pilot qualification requirements — flight time minimums, recency requirements, check ride currency — are being adapted for eVTOL operations through FAA rulemaking that is still in progress. Understanding what a Part 135 operator needs to be certificated, how the pilot training requirements are being adapted for eVTOL, and what the regulatory gap between the aircraft's type certificate and the operator's AOC means for the commercial service launch timeline are the commercial operations regulatory competencies that AAM operations engineers and certification teams require.

Done signal: You can describe the key components of a Part 135 Air Carrier Certificate — the Operations Specifications (OpSpecs) that define the authorised operations, the required manuals (Operations Manual, Minimum Equipment List), the pilot qualification requirements (PIC minimums, instrument currency, proficiency check schedule), and the maintenance programme approval. You can explain the specific AAM adaptation challenge — why the existing Part 135 pilot training requirements for rotorcraft or fixed-wing do not directly map to eVTOL operation, and what the FAA's powered-lift pilot type rating process requires. You can describe the timeline sequence from type certificate to first commercial revenue service, including the operator certification steps.

4–5 hours

12. Understand the AAM noise certification process — the current ICAO standards gap and the emerging means of compliance

Skill: The acoustic certification challenge that has no established regulatory solution for eVTOL.

Why the field cares: ICAO Annex 16 Volume I contains noise certification standards for subsonic jet and turboprop aircraft (Chapters 2–14), helicopters (Chapter 8), and tiltrotors (Chapter 12). It does not contain noise certification standards for multirotor electric VTOL aircraft — the configuration that represents the majority of AAM developers' products. This means that every eVTOL noise certification in Europe and the US currently proceeds through a Special Condition or an Equivalent Level of Safety process, with the developer proposing a means of compliance (typically: measure the noise at specified approach, departure, and overflight conditions using the applicable metric; demonstrate compliance with a proposed limit derived from analysis of community impact data) and the authority accepting or negotiating it. The ICAO Committee on Aviation Environmental Protection (CAEP) Working Group 1 is developing a noise standard for novel aircraft types, but it is not expected to be finalized before 2028. Understanding the current regulatory gap, how the interim ELOS process works in practice, and what measurement methodology and noise metric the leading developers have proposed for their certification are the AAM noise certification competencies that acoustic engineers and certification teams require.

Done signal: You can explain why ICAO Annex 16 Volume I does not currently apply to multirotor eVTOL aircraft, and describe the ELOS process through which noise certification is currently being progressed for the leading eVTOL developers. You can describe what the proposed noise measurement approach looks like — the reference measurement points for approach, departure, and overflight conditions, the proposed metric (SENEL, or an A-weighted equivalent sound exposure level), and the proposed certification limit basis (community impact analysis, comparison to helicopter limits at equivalent performance). You can describe the CAEP WG1 work programme timeline for novel aircraft noise standards.

4–6 hours

13. Prepare for a cross-domain AAM systems problem under interview conditions — a failure case analysis or a design trade-off with cascading implications

Skill: Structured AAM systems thinking under time pressure.

Why the field cares: Mid-level and senior AAM interviews consistently include a cross-domain systems problem — a failure case or a design trade that requires reasoning across at least three engineering domains simultaneously. “The FAA informs you that the maximum certified approach angle for your vertiport at the proposed downtown Manhattan site is 8 degrees rather than the 12 degrees you designed for, due to obstacle limitation surface constraints. Walk through the implications for your aircraft’s approach procedure, the noise footprint on the surrounding streets, the required hover power at the approach gate, the battery reserve margin on arrival, and the operator’s minimum fuel reserve policy.” The candidate who can trace these cascading implications — approach angle → approach speed → hover to land transition distance → noise footprint spatial extent → required approach power → battery consumption increase → reserve margin reduction → operator policy revision — is demonstrating the connected systems reasoning that AAM engineering requires.

Done signal: You can work through a cross-domain AAM systems problem in under 20 minutes — tracing the engineering implications of a constraint change (a noise limit, a vertiport geometry constraint, a battery energy limit, or a UTM throughput limit) through at least three engineering domains (aircraft performance, community noise, battery system, airspace management, commercial operations). You have practised this for at least two different scenarios. You articulate each step explicitly and identify the key trade-off at each cascade.

SENIOR LEVEL (Principal Engineer, Head of Certification, Chief Architect)

All mid-level tasks still apply. Regulatory knowledge must extend to direct experience with novel means of compliance development. Systems integration knowledge must include direct experience with programme-level certification strategy.

4–6 hours

14. Prepare to walk through the most technically challenging certification finding you have personally led — the means of compliance development and the authority engagement

Skill: Novel certification means of compliance development at the level a senior AAM certification engineer must demonstrate.

Why the field cares: Senior AAM certification engineers are specifically valued for having developed novel means of compliance — for having identified that an SC-VTOL-01 requirement could not be demonstrated by any standard method, designed an alternative demonstration approach, assembled the technical evidence package, and engaged with the EASA or FAA technical specialist until the approach was accepted. This experience is genuinely rare — most certification engineers work within established frameworks — and it is specifically the experience that AAM certification programmes need from their most senior technical personnel. The story must demonstrate not just that a novel means of compliance was needed, but that the candidate personally drove its development and acceptance.

Done signal: For one certification finding you personally led — where the standard means of compliance did not apply and a novel approach was required — you can describe: the specific SC-VTOL-01 or FAA Special Condition requirement, why the standard means of compliance was not applicable to your configuration, the alternative approach you proposed (the physical reasoning, the test methodology, the analysis supporting the alternative), the authority's initial position, the technical engagement through which the authority's concerns were addressed, and the form of the acceptance (an ELOS finding, an AMC development, or a Special Condition modification). You can describe the timeline of the engagement and the specific technical arguments that were most influential.

4–5 hours

15. Prepare a technically grounded position on one active AAM frontier — autonomous operation certification, vertiport energy management, or urban airspace architecture at high density

Skill: Forward-looking technical awareness at senior engineering depth.

Why the field cares: AAM in 2025–2026 has several genuine technical frontiers where the engineering challenges are real, the regulatory frameworks are being actively developed, and the architectural decisions being made now will determine what the industry looks like in 2030. Autonomous eVTOL operation — eliminating the pilot from the commercial operation — requires both a technical solution (an autonomous flight management system that meets or exceeds the safety provided by a certificated pilot in command) and a regulatory pathway (a certification basis for autonomous passenger transport that does not currently exist in Part 135 or EASA Air Ops). High-density urban airspace — dozens of aircraft simultaneously approaching and departing from adjacent vertiports — requires UTM algorithms that provide separation assurance at latencies compatible with the aircraft's response time, a challenge that is not solved by current UTM demonstrations that handle tens rather than thousands of simultaneous operations. Vertiport energy management — optimising the charging schedule across a fleet of arriving and departing aircraft against a grid connection constraint — is a real-time optimisation problem that has significant commercial implications but no established engineering solution. Senior engineers who can engage at technical depth with any of these frontiers are demonstrating the forward-looking perspective that the sector's senior roles require.

Done signal: You have a prepared 3-minute technical discussion covering one specific engineering challenge on one frontier — named specifically: the certification basis gap for autonomous passenger transport and what specific technical evidence would be required to close it; the separation assurance algorithm design for a vertiport complex handling 60 aircraft per hour and what response time constraint this places on the conflict resolution system; or the vertiport battery charging optimisation problem under a 2 MW grid connection constraint with stochastic arrival times and variable charge requirements. Your position identifies the specific unresolved engineering problem and describes the approaches being taken to address it.

4–5 hours

16. Prepare to lead a technical discussion on an AAM programme architecture decision — pilot-optional versus autonomous, single-FATO versus multi-FATO vertiport, or specific-aircraft-type

selection

4–5 hours

Skill: Strategic technical decision-making at the programme level.

Why the field cares: Senior AAM engineers are called on to make or advise on programme architecture decisions that have decade-scale commercial and technical consequences. “For a planned AAM operator launching in Singapore in 2027, recommend whether the initial fleet should be pilot-on-board or remote-piloted, given the current regulatory environment, the available certified aircraft types, the local pilot training supply, and the commercial economics.” This requires reasoning through: the regulatory availability of remote-piloted passenger operations in Singapore’s regulatory framework, what the pilot availability and training cost means for unit economics at the proposed fleet size, what the certification timeline difference between piloted and remote-piloted operation is, and what the public acceptance implication of each is in the local cultural context. The senior interview tests whether you can structure this reasoning and arrive at a specific recommendation.

Done signal: You have worked through at least two AAM programme architecture trade-off problems — one involving a pilot policy decision (pilot on board, single pilot, remote pilot, or autonomous) and one involving an infrastructure decision (vertiport site selection, FATO geometry, or charging infrastructure configuration). For each, you state the governing constraint, the two or three credible options, the key trade-off, and your recommendation with explicit sensitivities to the assumptions. You can do this in under 10 minutes per scenario.

17. Understand the autonomous flight management system — the technical requirements for safe autonomous operation and the current regulatory gap

4–5 hours

Skill: The technical and regulatory challenge that determines whether AAM’s economic model is achievable.

Why the field cares: Every current eVTOL certification programme is certifying with a pilot on board — because the regulatory framework for fully autonomous commercial passenger transport does not exist. The pilot cost in AAM operations (approximately \$60,000–120,000 per year in the US, more in Europe) is one of the largest operating cost items and is a significant contributor to the high initial fare prices that limit the addressable market. Autonomous operation — technically and regulatorily — is therefore the key to the mass-market AAM economics that justify the industry’s investment. The technical requirements for a safe autonomous flight management system (AFMS) include: a detect-and-avoid capability meeting the ASTM F3269 means of compliance for BVLOS operations, a failure mode management capability that handles all credible failure combinations without pilot intervention, a certification basis (DAL assignment, AFMS functional requirements, and verification methodology) that satisfies the authority’s requirement for equivalent safety to a certificated pilot, and an emergency landing capability that selects and executes an emergency landing in a survivable location without pilot command. None of these technical requirements is fully solved, and the regulatory framework for accepting them does not exist.

Done signal: You can describe the technical requirements for an autonomous flight management system at the level of its functional requirements — the specific functions it must perform (navigation, guidance, detect-and-avoid, system health monitoring, emergency management, passenger management) and the safety assurance level required for each. You can explain what ASTM F3269 requires for detect-and-avoid capability, how it would apply to a vertiport approach environment with multiple aircraft, and what the technology maturity gap is between current sense-and-avoid systems and this requirement. You can describe the regulatory gap — why no existing US or European regulation provides a pathway for fully autonomous commercial passenger transport — and what the industry’s engagement with regulators on this topic looks like.

18. Build quantitative estimation fluency for AAM system trade-offs — route economics, fleet sizing, vertiport throughput — without simulation

4–5 hours

Skill: Senior-speed first-principles reasoning about AAM commercial viability.

Why the field cares: Senior AAM engineers are expected to sanity-check commercial projections

4–5 hours

from business development teams, estimate whether a proposed aircraft design is commercially viable at the proposed route economics before the detailed financial model is built, and recognise when a technology assumption (a specific battery specific energy, a specific charge time) changes the commercial case from viable to unviable. These estimation problems are tractable from first principles: estimating the seat-mile cost from the aircraft capital cost, depreciation, maintenance, charging cost, and pilot cost; estimating the vertiport throughput from the FATO count, the approach and departure time, and the ground time; estimating the fleet size required to serve a specified number of passengers per day on a route with a specified flight time. None of these requires a detailed financial model — they require the governing commercial relationships applied with appropriate assumptions.

Done signal: You have worked through five AAM commercial estimation problems covering at least two of: seat-mile cost from aircraft operating cost components at a specified utilisation rate; vertiport throughput from FATO count and turnaround cycle time; fleet size for a specified passenger demand at a specified utilisation; battery energy cost per flight from electricity price and energy consumption; and route economics breakeven fare at a specified load factor. You state your assumptions before numbers, identify the most load-bearing commercial assumption, and give order-of-magnitude answers with stated uncertainty.

ALL LEVELS

1–2 hours

19. Prepare a specific answer to "Why AAM — and why this specific role and employer?" that reflects genuine technical and commercial engagement

Skill: Technically specific motivation that distinguishes AAM-committed candidates from those who find the concept exciting.

Why the field cares: AAM employers receive applications from candidates who are attracted by the excitement and novelty of the sector. What they are looking for is engineers who understand the specific technical challenges the sector faces and who have a specific engineering contribution to make to addressing them — not engineers who want to work on “the future of transportation.” The expected answer engages with one of: the SC-VTOL-01 certification challenge and how the candidate’s specific background applies to it; the UTM separation assurance problem and what algorithm approach the candidate would bring; the vertiport energy management optimisation and what the candidate’s control systems background enables; or the eVTOL handling qualities methodology gap and how the candidate’s flight dynamics research applies. The answer must name one specific engineering challenge, explain what makes it technically hard, and connect the candidate’s specific background to contributing to its solution — not just to the general mission.

Done signal: You have a 90-second answer that names the specific AAM technical challenge most relevant to your background, explains what makes it hard (the specific gap in current methodology, physics, or regulation), and connects your engineering background to contributing to its resolution. The answer demonstrates that you have researched the specific employer’s current programme status and can describe one specific certification, operational, or technical challenge they are currently facing that your background is relevant to. The answer cannot be given unchanged to a candidate applying for any other AAM employer.

1–2 hours

20. Map your technical background to a specific AAM employer type, role level, and technical domain — and understand where in the development cycle each employer is

Skill: Precise self-positioning in a sector where employer development stage matters more than in most aerospace disciplines.

Why the field cares: The AAM sector in 2025 spans employers at completely different points in their development cycle, and the engineering work — and the candidate characteristics required — differs dramatically between them. Joby Aviation is in active FAA certification review, finalising means of compliance and preparing for type inspection authorization. Archer Aviation is similarly advanced in its FAA engagement. Volocopter is pursuing EASA certification under SC-VTOL-01. Beta Tech-

1–2 hours

nologies is certifying its ALIA fixed-wing eSTOL alongside its eVTOL variant. Eve Air Mobility and Supernal are in preliminary design. Newer entrants are at concept development stage. A candidate targeting a certification role at Joby is applying to a programme that is making final certification arguments; a candidate targeting a design role at a later-stage entrant is applying to a programme that is making fundamental architectural decisions. The interview preparation, the relevant experience, and the career trajectory differ substantially between these. A candidate who has not understood where each employer is in its development cycle is not ready for the interview.

Done signal: You can state in under 60 seconds: the AAM employer you are targeting and its current development stage (concept, preliminary design, detailed design, certification, or operations), the specific technical role you are targeting and why your background matches the stage the employer is at, the specific SC-VTOL-01 or FAA powered-lift provisions most relevant to your target role, the simulation tools and analysis methods you work in fluently, and one specific technical challenge the employer is currently facing (derived from publicly available information — press releases, conference presentations, regulatory engagement documents) that your background is positioned to help with.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	AAM ecosystem — five components, constraints, and inter-system interactions	All	4–5 h
2	SC-VTOL-01 and FAA powered-lift — specific requirements, ELOS process, read the document	All	5–6 h
3	Power required curve — hover, transition hump, cruise, and what determines each	All	4–5 h
4	STAR stories — systems thinking, novel methodology, regulatory ambiguity, learning from failure	All	2–3 h
5	Urban noise and community acceptance — BVI, tonal noise, metrics, design approaches	All	3–4 h
6	Read five recent technical papers on one specific AAM challenge relevant to your discipline	All	6–8 h
7	DEP safety case — FMEA structure, independence assumptions, SC-VTOL-01 controllability	Mid+	5–6 h
8	Vertiport design requirements — FATO, obstacle limitation, charging infrastructure, throughput	Mid+	4–5 h
9	Novel methodology story — gap in standard approach, alternative developed, validation	Mid+	2 h
10	UAM airspace architecture — U-Space service layers, UTM, conflict management	Mid+	4–5 h
11	Part 135 commercial operations — operator certification, pilot qualification, AAM adaptation	Mid+	4–5 h
12	AAM noise certification — ICAO gap, ELOS process, SENEL metric, CAEP WG1	Mid+	4–5 h
13	Cross-domain AAM systems problem — cascading constraint analysis under time pressure	Mid+	4–6 h
14	Novel means of compliance story — regulatory gap, alternative approach, authority engagement	Senior	4–6 h
15	One active AAM frontier — autonomous operation, high-density airspace, or vertiport energy	Senior	4–5 h
16	AAM programme architecture decision — pilot policy or infrastructure configuration	Senior	4–5 h
17	Autonomous flight management system — technical requirements and regulatory gap	Senior	4–5 h
18	Quantitative estimation — seat-mile cost, vertiport throughput, fleet size, route economics	Senior	4–5 h
19	“Why AAM — why this employer and role?” — technically and commercially specific	All	1–2 h
20	Employer development stage / technical domain / SC-VTOL provisions self-positioning	All	1–2 h

THE STRUCTURAL REALITY THAT MAKES AAM DIFFERENT

Advanced Air Mobility is the only aerospace subdomain where the entire technology stack, the regulatory framework, the infrastructure, and the commercial model are all being developed simultaneously, each dependent on the others, none of them mature enough to define the others. The aircraft cannot be fully designed until the vertiport geometry is known. The vertiport cannot be sited until the noise profile is certified. The noise profile cannot be certified until the acoustic methodology is agreed with the authority. The acoustic methodology cannot be agreed until the aircraft's noise signature has been measured in flight test. The flight test cannot happen until the aircraft has a sufficient experimental airworthiness approval. And the experimental airworthiness approval depends on demonstrating that the distributed electric propulsion system is safe enough for test crew to fly — which requires a preliminary safety case based on a failure mode analysis of a system whose failure modes have not yet been observed in sufficient quantity to estimate failure rates from service data.

This circular dependency — in which every element of the system requires all other elements to be sufficiently mature before it can be completed — is the structural engineering challenge that defines AAM, and it is what makes the sector genuinely different from every other aerospace programme. In conventional aviation, the aircraft is designed, the aircraft is certified, and then airlines begin operating the aircraft. The regulatory framework is established. The infrastructure (airports) exists. The commercial model is proven. Each element of the system can draw on the maturity of the others.

In AAM, the engineers working in the sector are simultaneously building all of these elements — and managing the circular dependencies between them — with investors watching the clock and regulators watching the safety argument. The engineers who succeed in this environment are those who can hold the circular dependencies explicitly in mind — who always know which element they are currently dependent on, which element is dependent on them, and what the critical path looks like when their element is the bottleneck. They are also those who are genuinely comfortable making engineering decisions under uncertainty: choosing an aircraft configuration before the vertiport geometry is finalised, specifying a battery life limit before the degradation model is fully validated, proposing a means of compliance before the authority has expressed a position on it.

The opportunity in this environment — for engineers who can navigate the combination of technical rigor and systemic uncertainty — is genuine and historically unusual. The decisions being made in AAM engineering offices in 2025 will determine the shape of urban transportation in 2035, and the engineers making them will have the satisfaction of having built, from nothing, a new system of mobility. This is the specific appeal of the sector, and it is what every serious AAM employer is trying to find in the candidates they hire: engineers who understand what they are building, why it is difficult, and why that difficulty is, specifically, the reason they want to be there.